

MAM · Sample Proportions and Confidence Intervals —

Question Patterns and Solution Strategies

Basis: 29 past exam questions from 2016–2025, marking keys + exam reports (almost every year the reports call out "can calculate but can't explain"). Original questions are in [topics/07_Sample_Proportions_CI.pdf](#). The numerical examples in this document have been checked with Python. Review status: has been through one round of independent review and revised in line with the comments (see [Review_log.md](#)).

Formula card

- $\hat{p} = x/n$; the approximate distribution of $\hat{p}$: $\hat{p} \sim N(p, p(1-p)/n)$ (when n is large enough).
- Confidence interval: $\hat{p} \pm z \cdot \sqrt{\hat{p}(1-\hat{p})/n}$; margin of error $E = z \cdot \sqrt{\hat{p}(1-\hat{p})/n}$; interval width = $2E$.
- z values: 90% $\rightarrow$ 1.645, 95% $\rightarrow$ 1.960, 99% $\rightarrow$ 2.576.
- Reversing from an interval: $\hat{p} = (\text{upper bound} + \text{lower bound})/2$, $E = (\text{upper bound} - \text{lower bound})/2$.
- Sample size: $n = z^2 \hat{p}(1-\hat{p})/E^2$. For a minimum sample size (minimum / guarantee / ensure margin at most ...) always use $\hat{p} = 0.5$ (worst case), even if $\hat{p}$ has already been calculated earlier (the keys for 2023A-Q7(c), 2024A-Q10(f) and 2025A-Q16(e) all do this; the 2023 report specifically reminds you of this). Only substitute a particular sample proportion when the question explicitly asks you to (2016A-Q10(d): "Using the sample proportion of the survey...").

Markers' reminders (the reports say the most about this chapter)

1. When drawing a conclusion from an interval you must take a position (2023 report): you cannot dodge the conclusion with "cannot be 100% certain". Example (2024A-Q11(b)): "The claimed value 0.74 lies inside the 95% CI (0.652, 0.748), so there is insufficient evidence to reject the claim."
2. Bias questions must be specific (2022–2024 reports): state the source of the bias + explain which type of person is more or less likely to be sampled + link it to the context. Time and place are two different sources of bias; don't merge them into one point.
3. Sample sizes must be rounded up; when reversing a sample size, round to the nearest whole number (see Pattern 4 for the difference between the two).
4. Meaning of "95% confidence": if sampling were repeated many times, about 95% of the intervals would contain the true p ; any one specific interval either contains p or it doesn't.

Pattern 1: Write down the distribution of $\hat{p}$ and find a probability

Identify: "What is the (approximate) distribution of the sample proportion?", " $P(\hat{p} < 0.58)$ ".

Approach (3-mark template): ✓ "approximately normal (since n is large)" ✓ mean = p ✓ variance $p(1-p)/n$ (or write the standard deviation). Then calculate using normCdf, writing down the probability expression.

When the normal approximation cannot be used: the distribution graph is clearly right-skewed, n is small, np is too small (2023A-Q10(a): $n = 36$, $p = 0.05$, the graph is not symmetric $\Rightarrow$ not appropriate). For (a), writing "not symmetric" as the reason is enough. In (b) it changes to $n = 500$ and asks about the number of people X , so use the binomial; in (c), at $n = 500$ the normal approximation can be used again. What happens when the sample size increases: the standard deviation of $\hat{p}$ becomes smaller and the distribution becomes more concentrated, so $P(\hat{p} \leq 0.03)$ will increase (2017A-Q18(d)(ii): the two marks are given for "SD decreases" and "the distribution is more concentrated $\Rightarrow$ the probability is higher" respectively).

Past questions: 2016A-Q14(b), 2017A-Q18(d), 2018A-Q17(a)(b), 2019A-Q13(a), 2020A-Q12(b)(c), 2021A-Q11(a)(b), 2022A-Q13(a), 2023A-Q10, 2024A-Q10(b), 2025A-Q10(d).

Pattern 2: Calculate a confidence interval and margin of error

Approach: $\hat{p} \rightarrow$ choose the correct $z \rightarrow$ substitute into the formula (or use the CAS 1-Prop z Interval) $\rightarrow$ round to the required number of decimal places. Example (2023A-Q7): $76/200 = 0.38$, 95% CI $\approx$ (0.3127, 0.4473). Assumption: $\hat{p}$ is approximately normal (n large enough) — the key for 2016A-Q10(a) requires this

to be stated; random sampling can be added as a supplementary point.

Past questions: 2016A-Q10, 2017A-Q18(b)(c), 2018A-Q17(e), 2019A-Q8, 2020A-Q12(d)(e), 2021A-Q11(d)(e), 2023A-Q7(b), 2024A-Q9(a), 2025A-Q16(b)(c).

Pattern 3: Use a confidence interval to test a claim ★ appears every year

The standard two sentences (2 marks in the key)

1. "The claimed proportion $p = 0.6$ is **not** within the 95% confidence interval (0.4562, 0.5438)."
2. "Therefore there is sufficient evidence (at the 95% level) to conclude that the claim is incorrect / that Tina is unlikely to be correct."

Conversely, if the claimed value lies within the interval: "not enough evidence to conclude that the proportion has changed". **Comparing two intervals** (2020A-Q12(f)): the two intervals overlap $\Rightarrow$ there is insufficient evidence to say that the financial advice reduced the failure rate. **Why the true value not being in the interval doesn't necessarily mean a calculation error** (2019A-Q8(b), 2021A-Q13(g)): "Not all 95% confidence intervals contain the true proportion; about 5% will not, and this may be one of them."

Distinguishing the two ways a question can be asked (2025 report: statistical significance, as distinct from proving a result):

- If asked "is it different / what does it suggest" $\Rightarrow$ take a position using the two sentences above.
- If asked "does it **prove** the claim is incorrect / was a mistake made" $\Rightarrow$ answer **No**: an interval cannot prove anything, not all intervals contain p , and we don't know whether this interval is one of the ones that doesn't (2025A-Q15(f)).

Binomial distribution assumption questions (2020A-Q12(h) 4 marks, 2021A-Q11(g) 3 marks; the 2020 report specifically calls out poor answers): state the assumptions (each item is independent, the probability p is the same and constant) $\rightarrow$ judge whether they are reasonable in context $\rightarrow$ give a reason.

Past questions: 2016A-Q20(f), 2017A-Q12(e), 2018A-Q17(f), 2020A-Q12(f), 2021A-Q11(f), 2022A-Q12(g), 2023A-Q12(e), 2023A-Q13(d), 2024A-Q10(d), 2024A-Q11(b), 2025A-Q16(f).

Pattern 4: Sample size / reversing from an interval ★

A. Finding the minimum sample size (round up)

- "within 0.01 with 95% confidence" $\Rightarrow n = 1.96^2 \times 0.25 / 0.01^2 = 9604$ (using the exact z from the CAS gives 9603.6, which when rounded up is still **9604**) (2019A-Q14(a)).
- 2023A-Q7(c): $\hat{p} = 0.38$ has already been calculated earlier, but for "at least how many birds" still use $0.5 \Rightarrow n = 2400.9 \Rightarrow$ **2401** (substituting 0.38 gives 2263, losing marks).
- 2018A-Q13(b): 99%, $E = 0.08 \Rightarrow n = 259.2 \Rightarrow$ **260** (the key's exact words: rounds up to 260).
- 2018A-Q13(a) requires **using calculus to prove that the error is greatest when $\hat{p} = 0.5$** : $E \propto \sqrt{(\hat{p}(1-\hat{p}))}$; differentiate and set it to zero $\Rightarrow \hat{p} = 0.5$, then use the second derivative or a sign table to confirm it is a maximum.

B. Reversing $\hat{p}$, E , n and the number of people from an interval (round to the nearest whole number)

- 2024A-Q11: $\hat{p} = 0.70$, width 0.096 $\Rightarrow E = 0.048 \Rightarrow n = 1.96^2 \times 0.21 / 0.048^2 = 350.1 \Rightarrow$ **350**.
- 2018A-Q13(c): (0.342, 0.558) $\Rightarrow \hat{p} = 0.45$, $E = 0.108 \Rightarrow n \approx 140.8 \Rightarrow 141$, number of people $\approx 0.45 \times 141 \approx 63$.
- 2022A-Q13(d), 2023A-Q12(c)(d): given the interval and n , reverse to find the confidence level: first find E , then find $z = E/SE$, then get the confidence level from z (e.g. $z = 1.645 \Rightarrow 90\%$).

C. Proportional relationships (CF): $E \propto 1/\sqrt{n} \Rightarrow$ halving the error requires $n \times 4$ (2017F-Q4); to reduce the width to $1/3$ requires $n \times 9$ (2018F-Q5: 200 $\rightarrow$ 1800).

Past questions: 2016A-Q10(d), 2017F-Q4, 2018A-Q13, 2018F-Q5, 2019A-Q14, 2020A-Q14, 2022A-Q12(f), 2022A-Q13(d), 2023A-Q7(c), 2024A-Q10(f), 2024A-Q11(a), 2025A-Q16(e).

Pattern 5: Factors affecting interval width

Change	Interval width	Reason
n increases	Narrower	$SE = \sqrt{\hat{p}(1-\hat{p})/n}$ gets smaller
Confidence level increases (90% → 99%)	Wider	z gets larger
$\hat{p}$ closer to 0.5	Wider	$\hat{p}(1-\hat{p})$ gets larger (2025A-Q16(d)(iii): $\hat{p}$ further from 0.5 $\Rightarrow$ E decreases)

When answering, write both “direction + reason” (2019A-Q14(b) 4 marks: each of the two factors has 1 mark for “factor” and 1 mark for “effect”). If n is quadrupled, the width halves (2021A-Q13(f)).

Past exam questions: 2016A-Q14(c), 2019A-Q14(b), 2020A-Q12(g), 2021A-Q13(d)(f), 2022A-Q12(e), 2023A-Q12(f), 2024A-Q10(e), 2024A-Q11(c), 2025A-Q16(d).

Pattern 6: Sampling bias ★ present in most years

Answer template (usually 1–2 marks per source: identify the source + explain)

1. **Identify the source:** location (only outside the shopping centre/library), time (weekday mornings, lunchtime), self-selection (voluntary response), wording (leading question, e.g. “or an inferior brand?”), sampling frame (only mobile phone numbers, only sampling referred medical records), convenience (the first 200 copies printed, the newest printing machine).
2. **Explain which types of people are more or less likely to be sampled**, and link this to the context: time and location must be written as two separate points (2022 report). Example (2020A-Q14(c)):
 - Location: “Surveying outside the library over-represents people who already use the library, so $\hat{p}$ would overestimate the proportion of residents who use it.”
 - Time: “Surveying only at lunchtime under-represents residents who work or study during the day and cannot be there then.”

How to improve: randomly sample across the whole population (the whole week, all four printing machines); use random numbers to select times and subjects.

Past exam questions: 2017A-Q12(a), 2018A-Q17(c), 2019A-Q13(b)(c), 2020A-Q14(c), 2022A-Q13(b), 2023A-Q7(d), 2024A-Q10(a), 2025A-Q16(a).

Pattern 7: Binomial distribution of the number of intervals

Identify: “20 students each compute a 90% CI. X = number of intervals containing the true proportion.”

Approach: $X \sim \text{Bin}(20, 0.9)$, $E(X) = 18$, $\text{Var}(X) = 1.8$; “3 intervals do not contain p” $\Leftrightarrow X = 17$ (the key for 2024A-Q9(d) awards 1 mark for this conversion). **Past exam questions:** 2016A-Q20(d), 2024A-Q9(b)–(d).